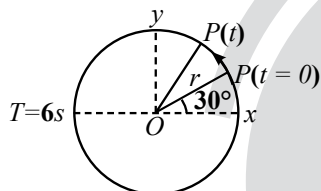


Simple Harmonic Motion

Single Correct Type Questions

1. For particle P revolving round the centre O with radius of circular path r and angular velocity ω , as shown in below figure, the projection of OP on the x -axis at time t is

[8 April, 2023 (Shift-II)]



- (a) $x(t) = r \cos\left(\omega t + \frac{\pi}{6}\right)$ (b) $x(t) = r \cos(\omega t)$
 (c) $x(t) = r \sin\left(\omega t + \frac{\pi}{6}\right)$ (d) $x(t) = r \cos\left(\omega t - \frac{\pi}{6}\right)$
2. A particle executes simple harmonic motion between $x = -A$ and $x = +A$. If time taken by the particle to go from $x = 0$ to $\frac{A}{2}$ is $2s$; then time taken by the particle in going from $x = \frac{A}{2}$ to A will be:

[25 Jan, 2023 (Shift-II)]

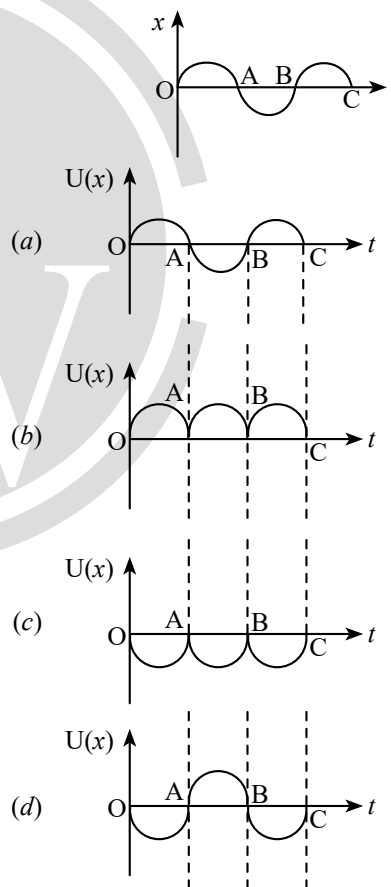
- (a) $3s$ (b) $2s$
 (c) $1.5s$ (d) $4s$
3. For a periodic motion represented by the equation $y = \sin \omega t + \cos \omega t$. The amplitude of the motion is

[10 April, 2023 (Shift-II)]

- (a) 0.5
 (b) $\sqrt{2}$
 (c) 1
 (d) 2

4. The variation of displacement with time of a particle executing free simple harmonic motion is shown in the figure. The potential energy $U(x)$ versus time (t) plot of the particle is correctly shown in figure:

[27 Aug, 2021 (Shift-I)]



5. Assume that a tunnel is dug along a chord of the earth, at a perpendicular distance ($R/2$) from the earth's centre, where ' R ' is the radius of the Earth. The wall of the tunnel is frictionless. If a particle is released in this tunnel, it will execute a simple harmonic motion with a time period :

[26 Feb, 2021 (Shift-I)]

$$(a) 2\pi\sqrt{\frac{R}{g}} \quad (b) \frac{g}{2\pi R}$$

$$(c) \frac{1}{2p}\sqrt{\frac{g}{R}} \quad (d) \frac{2pR}{g}$$

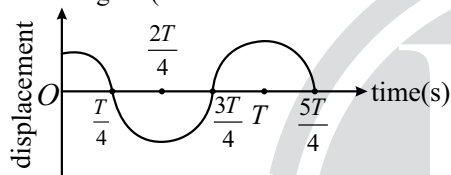
6. A particle is making simple harmonic motion along the X-axis. If at distances x_1 and x_2 from the mean position the velocities of the particle are v_1 and v_2 respectively. The time period of its oscillation is given as:

[20 July, 2021 (Shift-II)]

$$(a) T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}} \quad (b) T = 2\pi\sqrt{\frac{x_2^2 + x_1^2}{v_1^2 - v_2^2}}$$

$$(c) T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 + v_2^2}} \quad (d) T = 2\pi\sqrt{\frac{x_2^2 + x_1^2}{v_1^2 + v_2^2}}$$

7. The displacement time graph of a particle executing S.H.M is given in figure (sketch is schematic and not to scale)



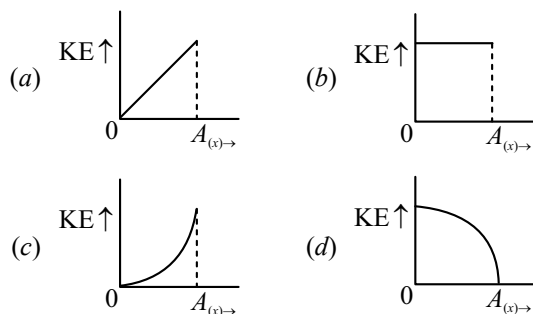
Which of the following statements is/are true for this motion?

[2 Sep, 2020 (Shift-II)]

- A. The force is zero at $t = \frac{3T}{4}$
 B. The acceleration is maximum at $t = T$
 C. The speed is maximum at $t = \frac{T}{4}$
 D. The P.E. is equal to K.E. of the oscillation at $t = \frac{T}{2}$
- (a) B, C and D (b) A and D
 (c) A, B and C (d) A, B and D

8. The variation of kinetic energy (KE) of a particle executing simple harmonic motion with the displacement (x) starting from mean position to extreme position (A) is given by (1)

[11 April, 2023 (Shift-I)]



9. For a body executing S.H.M.:

- (A) Potential energy is always equal to its K.E.
 (B) Average potential and kinetic energy over any given time interval are always equal.
 (C) Sum of the kinetic and potential energy at any point of time is constant.
 (D) Average K.E. in one time period is equal to average potential energy in one time period.

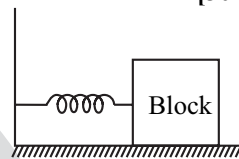
Choose the most appropriate option from the options given below:

[31 Aug, 2021 (Shift-II)]

- (a) Only (B) (b) (B) and (C)
 (c) Only (C) (d) (C) and (D)

10. For a simple harmonic motion in a mass spring system shown, the surface is frictionless. When the mass of the block is 1kg, the angular frequency is ω_1 . When the mass of the block is 2 kg the angular frequency is ω_2 . The ratio ω_2/ω_1 is:

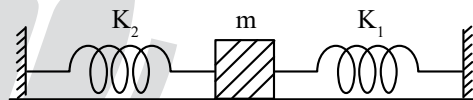
[30 Jan, 2023 (Shift-II)]



- (a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) 2 (d) $\frac{1}{2}$

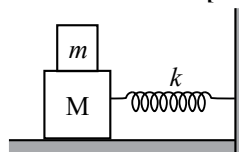
11. A mass m is attached to two springs as shown in figure. The spring constants of two springs are K_1 and K_2 . For the frictionless surface, the time period of oscillation of mass m is

[06 April, 2023 (Shift-I)]



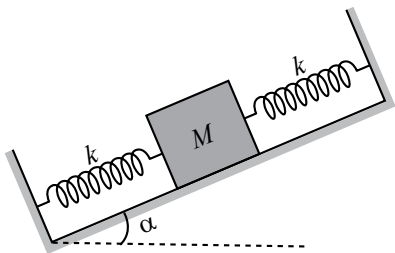
- (a) $\frac{1}{2\pi}\sqrt{\frac{K_1 + K_2}{m}}$ (b) $\frac{1}{2\pi}\sqrt{\frac{K_1 - K_2}{m}}$
 (c) $2\pi\sqrt{\frac{m}{K_1 + K_2}}$ (d) $2\pi\sqrt{\frac{m}{K_1 - K_2}}$

12. In the given figure, a mass M is attached to a horizontal spring which is fixed on one side to a rigid support. The spring constant of the spring is k . The mass oscillates on a frictionless surface with time period T and amplitude A . When the mass is in equilibrium position, as shown in the figure, another mass m is gently fixed upon it. New amplitude of oscillation will be: [24 Feb, 2021 (Shift-I)]

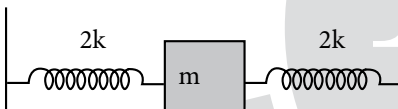


- (a) $A\sqrt{\frac{M}{M-m}}$ (b) $A\sqrt{\frac{M}{M+m}}$
 (c) $A\sqrt{\frac{M-m}{M}}$ (d) $A\sqrt{\frac{M+m}{M}}$

13. In the given figure, a body of mass M is held between two massless springs, on a smooth inclined plane. The free ends of the springs are attached to firm supports. If each spring has spring constant k , the frequency of oscillation of given body is: [24 Feb, 2021 (Shift-II)]



- (a) $\frac{1}{2\pi} \sqrt{\frac{k}{2m}}$ (b) $\frac{1}{2\pi} \sqrt{\frac{k}{Mg \sin \alpha}}$
 (c) $\frac{1}{2\pi} \sqrt{\frac{2k}{M}}$ (d) $\frac{1}{2\pi} \sqrt{\frac{2k}{Mg \sin \alpha}}$
14. Two identical springs of spring constant ' $2k$ ' are attached to a block of mass m and to fixed support (see figure). When the mass is displaced from equilibrium position on either side, it executes simple harmonic motion. The time period of oscillations of this system is: [25 Feb, 2021 (Shift-II)]



- (a) $\pi \sqrt{\frac{m}{k}}$ (b) $\pi \sqrt{\frac{m}{2k}}$
 (c) $2\pi \sqrt{\frac{m}{k}}$ (d) $2\pi \sqrt{\frac{m}{2k}}$
15. The time period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle, which moves without friction down an inclined plane of inclination α , is given by. [29 July, 2022 (Shift-I)]
- (a) $2\pi \sqrt{L/(g \cos \alpha)}$ (b) $2\pi \sqrt{L/(g \sin \alpha)}$
 (c) $2\pi \sqrt{L/g}$ (d) $2\pi \sqrt{L/(g \tan \alpha)}$
16. Time period of a simple pendulum is T inside a lift when the lift is stationary. If the lift moves upwards with an acceleration $g/2$, the time period of pendulum will be: [16 March, 2021 (Shift-I)]

- (a) $\sqrt{\frac{2}{3}}T$ (b) $\sqrt{3}T$
 (c) $\frac{T}{\sqrt{3}}$ (d) $\sqrt{\frac{3}{2}}T$

17. A simple pendulum, made of a string of length l and a bob of mass m , is released from a small angle θ_0 . It strikes a block of mass M , kept on a horizontal surface at its lowest point of oscillations, elastically. It bounces back and goes up to an angle θ_1 . Then M is given by: [12 Jan, 2019 (Shift-I)]

- (a) $\frac{m}{2} \left(\frac{\theta_0 + \theta_1}{\theta_0 - \theta_1} \right)$ (b) $m \left(\frac{\theta_0 - \theta_1}{\theta_0 + \theta_1} \right)$
 (c) $m \left(\frac{\theta_0 + \theta_1}{\theta_0 - \theta_1} \right)$ (d) $\frac{m}{2} \left(\frac{\theta_0 - \theta_1}{\theta_0 + \theta_1} \right)$

18. A simple pendulum oscillating in air has period T . The bob of the pendulum is completely immersed in a non-viscous liquid. The density of the liquid is $\frac{1}{16}$ th of the material of the bob. If the bob is inside liquid all the time, its period of oscillation in this liquid is: [9 April, 2019 (Shift-I)]

- (a) $4T \sqrt{\frac{1}{15}}$ (b) $2T \sqrt{\frac{1}{10}}$ (c) $4T \sqrt{\frac{1}{14}}$ (d) $2T \sqrt{\frac{1}{14}}$

19. The mass and the diameter of a planet are three times the respective values for the Earth. The period of oscillation of a simple pendulum on the Earth is 2s. The period of oscillation of the same pendulum on the planet would be: [11 Jan, 2019 (Shift-II)]

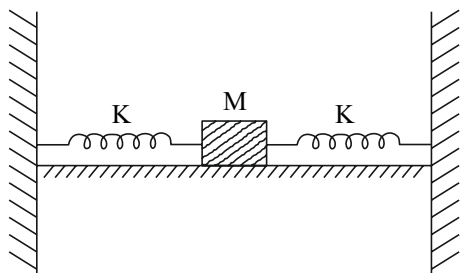
- (a) $\frac{\sqrt{3}}{2}s$ (b) $\frac{2}{\sqrt{3}}s$ (c) $\frac{3}{2}s$ (d) $2\sqrt{3}s$

Integer Type Questions

20. Two simple harmonic motion, are represented by the equations $y_1 = 10 \sin \left(3\pi t + \frac{\pi}{3} \right)$
 $y_2 = 5(\sin 3\pi t + \sqrt{3} \cos 3\pi t)$ Ratio of amplitude of y_1 to $y_2 = x : 1$. The value of x is ____ [27 Aug, 2021 (Shift-II)]
21. The general displacement of a simple harmonic oscillator is $x = A \sin \omega t$. Let T be its time period. The slope of its potential energy (U) - time (t) curve will be maximum when $t = \frac{T}{\beta}$. The value of β is ____ [30 Jan, 2023 (Shift-I)]
22. At a given point of time the value of displacement of a simple harmonic oscillator is given as $y = A \cos (30^\circ)$. If amplitude is 40 cm and kinetic energy at that time is 200 J, the value of force constant is $1.0 \times 10^x \text{ Nm}^{-1}$. The value of x is ____ [13 April, 2023 (Shift-I)]

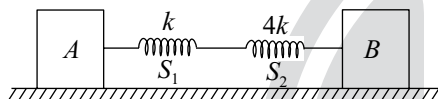
23. In the figure given below, a block of mass $M = 490 \text{ g}$ placed on a frictionless table is connected with two springs having same spring constant ($K = 2 \text{ N m}^{-1}$). If the block is displaced horizontally through ' x ' m then the number of complete oscillations it will make in 14π seconds will be _____.

[31 Jan, 2023 (Shift-I)]



24. In the reported figure, two bodies A and B of masses 200 g and 800 g are attached with the system of springs. Springs are kept in a stretched position with some extension when the system is released. The horizontal surface is assumed to be frictionless. The angular frequency will be rad/s when $k = 20 \text{ N/m}$.

[25 July, 2021 (Shift-I)]



25. Consider two identical springs each of spring constant k and negligible mass compared to the mass M as shown. Fig. 1 shows one of them and Fig. 2 shows their series combination. The ratios of time period of oscillation of the two SHM is

$$\frac{T_b}{T_a} = \sqrt{x}, \text{ Where value of } x \text{ is } \underline{\hspace{2cm}}.$$

[17 March, 2021 (Shift-I)]

(Round off to the Nearest Integer)

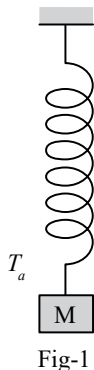


Fig-1

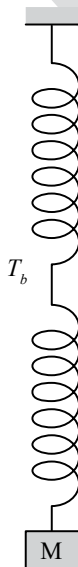
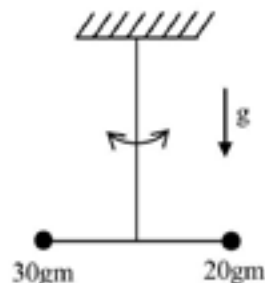


Fig-2

26. Two point-like objects of masses 20 gm and 30 gm are fixed at the two ends of a rigid massless rod of length 10 cm . This system is suspended vertically from a rigid ceiling using a thin wire attached to its center of mass, as shown in the figure. The resulting torsional pendulum undergoes small oscillations. The torsional constant of the wire is $1.2 \times 10^{-8} \text{ N m rad}^{-1}$. The angular frequency of the oscillations is $n \times 10^{-3} \text{ rad s}^{-1}$. The value of n is _____.

[JEE , 2023]



27. Time period of a simple pendulum is T . The time taken to complete $\frac{5}{8}$ oscillations starting from mean position is

$$\frac{\alpha}{\beta} T. \text{ The value of } \alpha \text{ is } \underline{\hspace{2cm}}. \text{ [26 Feb, 2021 (Shift-II)]}$$

28. A pendulum bob has a speed of 3 m/s at its lowest position. The pendulum is 50 cm long. The speed of bob, when the length makes an angle of 60° to the vertical will be ($g = 10 \text{ m/s}^2$) _____ m/s .

[25 July, 2021 (Shift-I)]

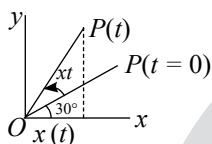
ANSWER KEY

1. (a) 2. (d) 3. (b) 4. (b) 5. (a) 6. (a) 7. (c) 8. (d) 9. (d) 10. (b)
 11. (c) 12. (b) 13. (c) 14. (a) 15. (a) 16. (a) 17. (c) 18. (a) 19. (d) 20. [1]
 21. [8] 22. [4] 23. [20] 24. [10] 25. [2] 26. [10] 27. [7] 28. [2]

EXPLANATIONS

1. (a) Given $\theta_1 = 30^\circ = \frac{\pi}{6}$

At time t angle swept by point $P, \theta_2 = \omega t$



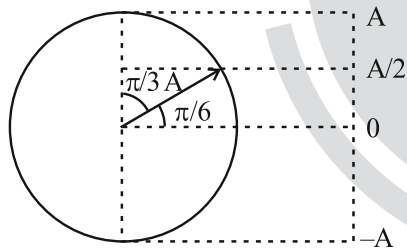
Projection on x -axis

$$x(t) = r \cos(\theta_1 + \theta_2)$$

$$x(t) = r \cos(\omega t + \pi/6)$$

$$\Rightarrow T \approx 1 \text{ hour } 24 \text{ minutes}$$

2. (d)



Let time taken by the particle to move from 0 to $\frac{A}{2}$ is t_1

& time taken by the particle to move from $\frac{A}{2}$ to A is t_2

$$\Rightarrow \omega t_1 = \frac{\pi}{6}$$

$$\text{and } \omega t_2 = \frac{\pi}{3}$$

$$\Rightarrow \frac{t_1}{t_2} = \frac{1}{2}$$

$$t_2 = 2t_1 = 2 \times 2 = 4 \text{ sec}$$

3. (b) $y = \sin \omega t + \cos \omega t$

$$y = \sin \omega t + \sin\left(\omega t + \frac{\pi}{2}\right)$$

$$\Delta\phi = \frac{\pi}{2} = 90^\circ$$

Amplitude of the motion,

$$A = \sqrt{1^2 + 1^2 + 2 \times 1 \times 1 \times \cos(90^\circ)} = \sqrt{2}$$

4. (b) We know that for SHM, potential energy is minimum of mean position and is maximum at its extreme position. Also, the sign of potential energy is the same throughout the SHM.

5. (a) Force, $F = m E \cos \theta$

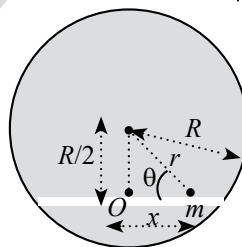
$$= m \left(\frac{GM}{R^3} \times r \right) \times \frac{x}{r}$$

$$= \frac{mgR^2}{R^3} \times x$$

$$\left[g = \frac{GM}{R^2} \right]$$

$$ma = \frac{mg}{R} x \Rightarrow a = \frac{g}{R} x$$

$$\text{Time period of SHM, } T = 2\pi \sqrt{\frac{R}{g}}$$



O $\Rightarrow$ Mean Position

6. (a) As we know that, $v^2 = \omega^2 (A^2 - x^2)$ for SHM

$$\therefore v_1^2 = \omega^2 (A^2 - x_1^2) \quad \dots(i)$$

$$v_2^2 = \omega^2 (A^2 - x_2^2) \quad \dots(ii)$$

Subtracting equation (ii) from equation (i), we get

$$v_1^2 - v_2^2 = \omega^2 (x_2^2 - x_1^2) \Rightarrow \omega = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}}$$

$$\Rightarrow T = \frac{2\pi}{\omega} \Rightarrow T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$

7. (c) At $t = \frac{3T}{4}$, particle is at mean position
 $\Rightarrow$ Force = 0
 At $t = T$, particle is at extreme position
 $\Rightarrow$ Acceleration is maximum
 At $t = \frac{T}{4}$, particle is at mean position
 $\Rightarrow$ Speed is maximum
 At $t = \frac{T}{2}$, particle is at extreme position
 $\Rightarrow$ Speed = 0 $\Rightarrow$ K.E. = 0 $\Rightarrow$ P.E. is maximum.

8. (d) Kinetic energy of particle executing SHM

$$\text{KE} = \frac{1}{2} m \omega^2 (A^2 - x^2) \quad \dots (i)$$

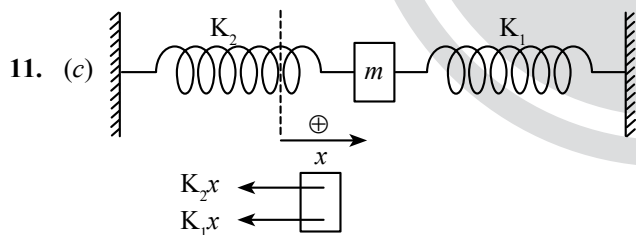
At $x = 0$, KE is maximum while at $x = A$, KE is zero
 Also from (i) KE vs x graph is parabola.

9. (d) In simple harmonic motion, sum of K.E. and P.E. remains constant and $\langle \text{K.E.} \rangle = \langle \text{P.E.} \rangle$ [for one time-period]

10. (b) Angular frequency ω is given by, $\omega = \sqrt{\frac{k}{m}}$

$$\Rightarrow \omega \propto \frac{1}{\sqrt{m}}$$

$$\Rightarrow \frac{\omega_2}{\omega_1} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{1}{2}}$$



On displacing m to right by x

$$F = -(K_1 x + K_2 x) = -(K_1 + K_2)x$$

$$a = \frac{F}{m} = -\left(\frac{K_1 + K_2}{m}\right)x = -\omega^2 x$$

$$\therefore \omega = \sqrt{\frac{K_1 + K_2}{m}} \Rightarrow T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{K_1 + K_2}}$$

12. (b) In first case, velocity of block in equilibrium

$$\text{condition} \quad v = A\omega = A \cdot \sqrt{\frac{k}{M}}$$

In second case, velocity of block in equilibrium

$$\text{condition, } v' = A'\omega' = A' \sqrt{\frac{k}{M+m}}$$

Using conservation of momentum, $Mv = (M+m)v'$

$$\Rightarrow MA \sqrt{\frac{k}{M}} = (M+m) A' \sqrt{\frac{k}{M+m}} \Rightarrow A' = A \sqrt{\frac{M}{M+m}}$$

13. (c) Since, $k_{eq} = 2k$

$$\therefore f = \frac{1}{2\pi} \sqrt{\frac{k_{eq}}{m}} = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$$

14. (a) Equivalent spring constant = $2k + 2k = 4k$

$$T = 2\pi \sqrt{\frac{m}{4k}} = \pi \sqrt{\frac{m}{k}}$$

15. (a) $g_{eff} = g \cos \alpha$, $T = 2\pi \sqrt{\frac{L}{g \cos \alpha}}$

16. (a) $T = 2\pi \sqrt{\frac{\ell}{g_{eff}}}$

When lift is stationary, then $g_{eff} = g$, so $T = 2\pi \sqrt{\frac{\ell}{g}}$

When lift is moving up with acceleration $g/2$, then

$$g_{eff} = \frac{3g}{2}, \text{ so } T = 2\pi \sqrt{\frac{2\ell}{3g}} = \sqrt{\frac{2}{3}} T$$

17. (c) **Case 1:** Before collision,

$$\text{Speed, } v = \sqrt{2gL(1 - \cos \theta_0)}$$

Case 2: After collision,

$$\text{Speed, } v_1 = \sqrt{2gL(1 - \cos \theta_1)}$$

$$\therefore \frac{v_1}{v} = \frac{M-m}{M+m}$$

$$\therefore \frac{\sqrt{1 - \cos \theta_1}}{\sqrt{1 - \cos \theta_0}} = \frac{M-m}{M+m}$$

$$\frac{\sin(\theta_1/2)}{\sin(\theta_0/2)} = \frac{M-m}{M+m} \quad \left[\because 1 - \cos 2\theta = 2 \sin^2 \theta \right]$$

$$\frac{\theta_1}{\theta_0} = \frac{M-m}{M+m} \quad [\text{As angle is small}]$$

$$M\theta_1 + m\theta_1 = M\theta_0 - m\theta_0$$

$$M(\theta_0 - \theta_1) = m(\theta_1 + \theta_0)$$

$$M = m \left[\frac{\theta_1 + \theta_0}{\theta_0 - \theta_1} \right]$$

18. (a) For a simple pendulum $T = 2\pi \sqrt{\frac{L}{g_{\text{eff}}}}$

Case 1: when pendulum is in air $\rightarrow g_{\text{eff}} = g$

Case 2: when pendulum is in liquid

$$g_{\text{eff}} = g \left(1 - \frac{\rho_{\text{liquid}}}{\rho_{\text{body}}} \right) = g \left(1 - \frac{1}{16} \right) = \frac{15g}{16}$$

$$\text{So, } \frac{T'}{T} = \frac{2\pi \sqrt{\frac{L}{15g/16}}}{2\pi \sqrt{\frac{L}{g}}} \Rightarrow T' = \frac{4T}{\sqrt{15}}$$

19. (d) $\because g = \frac{GM}{R^2}$

$$\frac{g_p}{g_e} = \frac{M_p}{M_e} \left(\frac{R_e}{R_p} \right)^2 = 3 \left(\frac{1}{3} \right)^2 = \frac{1}{3}$$

$$\text{Also, } T \propto \frac{1}{\sqrt{g}}$$

$$\Rightarrow \frac{T_p}{T_e} = \sqrt{\frac{g_e}{g_p}} = \sqrt{3} \Rightarrow T_p = 2\sqrt{3} \text{ s}$$

20. [1] It is given that, amplitude of $y_1 = A_1 = 10$

Therefore, amplitude of

$$y_2 = A_2 = \sqrt{(5)^2 + (5\sqrt{3})^2} = \sqrt{100} = 10$$

$$\Rightarrow \frac{A_1}{A_2} = \frac{10}{10} = 1$$

Hence, the correct value of x in $x : 1$ is 1.

21. [8] $x = A \sin(\omega t)$

$$U_{(x)} = \frac{1}{2} kx^2$$

$$\frac{dU}{dt} = \frac{1}{2} k 2x \frac{dx}{dt}$$

$$= kA^2 \omega \sin \omega t \cos \omega t \times \frac{2}{2}$$

$$\left(\frac{dU}{dt} \right)_{\text{max}} = \frac{kAA^2 \omega}{2} (\sin 2\omega t)_{\text{max}}$$

$$2\omega t = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{4} \omega = \frac{T}{8} \Rightarrow \beta = 8$$

22. [4] General equation, $x = A \sin(\omega t + \phi)$ at given time

$$\Rightarrow y = 40 \times \frac{\sqrt{3}}{2} = 20\sqrt{3} \text{ cm}$$

$$\Rightarrow A = 40 \text{ cm}$$

$$\Rightarrow K.E. = \frac{1}{2} k(A^2 - y^2) = 200$$

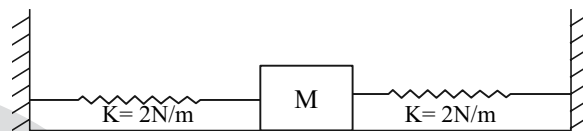
$$200 = \frac{1}{2} k \left(\frac{1600 - 1200}{100 \times 100} \right)$$

$$400 \times 100 \times 100 = k \times 400$$

$$k = 10^4$$

$$x = 4$$

23. [20]



$K_{\text{eff}} = K + K$ as both the springs are in parallel

$$= 2K = 4 \text{ N/m}$$

$$m = 0.49$$

$$\therefore T = 2\pi \sqrt{\frac{m}{K_{\text{eff}}}} = 2\pi \sqrt{\frac{0.49 \text{ kg}}{4}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{49}{400}} = \frac{7\pi}{10}$$

No. of oscillation in the 14π second is

$$N = \frac{\text{time}}{T} = \frac{14\pi}{7\pi/10} = 20$$

24. [10] Reduced mass of the system, $\mu = \frac{m_1 m_2}{m_1 + m_2}$

$$\mu = \frac{0.2 \times 0.8}{0.2 + 0.8} = 0.16 \text{ kg}$$

Equivalent spring constant,

$$K_{\text{eq}} = \frac{k \times 4k}{k + 4k} = \frac{4k}{5} = 4 \times \frac{20}{5} = 16$$

$$\omega = \sqrt{\frac{k_{\text{eq}}}{\mu}}$$

$$\omega = \sqrt{\frac{16}{0.16}} = 10 \text{ rad/s}$$

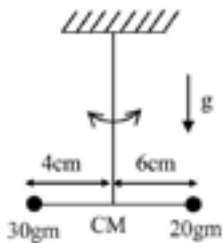
25. [2] For Figure 1, time period, $T_a = 2\pi \sqrt{\frac{m}{k}}$

$$\text{For Figure-2, } K_{\text{eq}} = \frac{K_1 \cdot K_2}{K_1 + K_2} = \frac{K}{2}$$

Therefore, time period, $T_b = 2\pi\sqrt{\frac{2m}{k}}$

$$\Rightarrow \frac{T_b}{T_a} = \sqrt{2}$$

26. [10]



For torsional pendulum, $T = 2\pi\sqrt{\frac{I}{C}}$

$$\text{or } \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{C}}$$

$$\therefore \omega = \sqrt{\frac{C}{I}}$$

Moment of inertia of the system,

$$I = (30)(4)^2 + (20)(6)^2 = 1200 \text{ gm-cm}^2 \\ = 1.2 \times 10^{-4} \text{ kg-m}^2$$

$$\therefore \omega = \sqrt{\frac{1.2 \times 10^{-8}}{1.2 \times 10^{-4}}}$$

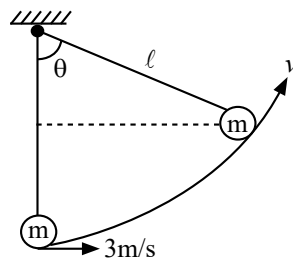
$$\omega = (10^{-2}) = 10 \times 10^{-3} \text{ rad/s, } n = 10$$

$$27. [7] \text{ Total oscillation} = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}$$

$$\text{Total time in oscillation} = \frac{T}{2} + \frac{T}{12} = \frac{7T}{12}$$

$$\therefore \frac{\alpha}{\beta} T = \frac{7}{12} T \Rightarrow \alpha = 7, \beta = 12$$

28. [2] From conservation of Energy



$$\frac{1}{2} mu^2 = \frac{1}{2} mv^2 + mg\ell(1 - \cos\theta)$$

$$\Rightarrow v^2 = u^2 - 2g\ell(1 - \cos\theta)$$

$$\Rightarrow v^2 = 3^2 - 2 \times 10 \times 0.5 \times \left(1 - \frac{1}{2}\right)$$

$$\Rightarrow v^2 = 4$$

$$v = 2 \text{ m/s}$$